

Cross Platform App Development Lab

Grocery Ordering App UI

Aim	Design and develop different UI in Flutter using Row, Column, and Stack widgets together.
Concept Used	Row for horizontal layout, Column for vertical layout, Stack for overlapping elements such as badges and floating buttons.

1. Objective

To apply Row, Column, and Stack widgets together in a single, realistic UI screen .

2. UI Elements and Layout Mapping

- App bar (delivery address + cart icon) — built using a Row with space-between alignment.
- Search bar — a Row containing a search icon and hint text.
- Category strip (Fruits, Dairy, Snacks, Bakery) — a horizontally scrollable Row of circular icons.
- Product grid — a Column of Rows, where each Row holds two product cards side by side.
- Product card — a Stack, with the product image at the base, a discount badge positioned top-left, and a favorite icon positioned top-right.
- Floating cart bar — a Stack placed over the whole screen, positioned near the bottom using Positioned, so it floats above the scrollable content.

3. Steps

1. Create a new Flutter project named zepto_ui_demo.
2. Open lib/main.dart and remove the default boilerplate code.
3. Build the app bar as a Row showing the delivery address on the left and a cart icon on the right.
4. Add a search bar below the app bar as a Row with a search icon and placeholder text.
5. Add a horizontally scrollable Row for category icons (Fruits, Dairy, Snacks, Bakery).
6. Create a reusable ProductCard widget using a Stack: base image container, a Positioned discount badge, and a Positioned favorite icon.
7. Arrange ProductCard widgets in pairs using Row, and stack those Rows vertically using a Column to form the product grid.
8. Wrap the whole scrollable content in a Stack along with a floating cart bar, and position the cart bar near the bottom using Positioned.
9. Save the file, connect a device or emulator, and run the application.
10. Scroll through the product grid and verify the floating cart bar stays fixed at the bottom.
11. Tap the ADD button on a product card (optional enhancement) and update the item count and total shown on the floating cart bar.
12. Take a screenshot of the running application for the lab record.

4. Sample Code

File: lib/main.dart

```
import 'package:flutter/material.dart';

void main() => runApp(const ZeptoUIDemo());

class ZeptoUIDemo extends StatelessWidget {
  const ZeptoUIDemo({super.key});
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      debugShowCheckedModeBanner: false,
      home: const HomeScreen(),
    );
  }
}

class HomeScreen extends StatelessWidget {
  const HomeScreen({super.key});

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      backgroundColor: Colors.white,
      // Outer Stack: scrollable content + floating cart bar
      body: Stack(
        children: [
          Column(
            children: [
              buildAppBar(),
              buildSearchBar(),
              buildCategoryRow(),
              const SizedBox(height: 8),
              Expanded(child: buildProductGrid()),
            ],
          ),
          Positioned(
            left: 14,
            right: 14,
            bottom: 10,
            child: buildFloatingCartBar(),
          ),
        ],
      ),
    );
  }

  // ---- App bar: Row ----
  Widget buildAppBar() {
    return Container(
      color: const Color(0xFF6A0DAD),
      padding: const EdgeInsets.all(14),
      child: Row(
        mainAxisAlignment: MainAxisAlignment.spaceBetween,
        children: const [
          Column(
            crossAxisAlignment: CrossAxisAlignment.start,
            children: [
              Text('Delivery in 9 minutes', style: TextStyle(color: Colors.white70,
                fontSize: 11)),
              Text('Home - 221B Baker Street',
```

```

        style: TextStyle(color: Colors.white, fontWeight: FontWeight.bold)),
      ],
    ),
    Icon(Icons.shopping_cart_outlined, color: Colors.white),
  ],
),
);
}

// ---- Search bar: Row ----
Widget buildSearchBar() {
  return Container(
    margin: const EdgeInsets.all(14),
    padding: const EdgeInsets.symmetric(horizontal: 12, vertical: 10),
    decoration: BoxDecoration(
      color: const Color(0xFFF0EEF5),
      borderRadius: BorderRadius.circular(10),
    ),
    child: const Row(
      children: [
        Icon(Icons.search, color: Colors.grey, size: 18),
        SizedBox(width: 8),
        Text('Search for atta, dal, oil...', style: TextStyle(color: Colors.grey)),
      ],
    ),
  );
}

// ---- Category strip: horizontally scrollable Row ----
Widget buildCategoryRow() {
  final categories = ['Fruits', 'Dairy', 'Snacks', 'Bakery'];
  return SizedBox(
    height: 80,
    child: ListView(
      scrollDirection: Axis.horizontal,
      padding: const EdgeInsets.symmetric(horizontal: 14),
      children: categories
        .map((c) => Padding(
          padding: const EdgeInsets.only(right: 10),
          child: Column(
            children: [
              CircleAvatar(radius: 25, backgroundColor: const
Color(0xFFF0EEF5)),
              const SizedBox(height: 4),
              Text(c, style: const TextStyle(fontSize: 10)),
            ],
          ),
        ))
        .toList(),
    ),
  );
}

// ---- Product grid: Column of Rows ----
Widget buildProductGrid() {
  return ListView(
    padding: const EdgeInsets.symmetric(horizontal: 14),
    children: const [
      Row(
        children: [

```

```

        Expanded(child: ProductCard(name: 'Fresh Bananas', price: 48, mrp: 60,
discount: '20% OFF')),
        SizedBox(width: 12),
        Expanded(child: ProductCard(name: 'Amul Milk', price: 27, mrp: 30, discount:
'10% OFF')),
      ],
    ),
    SizedBox(height: 12),
    Row(
      children: [
        Expanded(child: ProductCard(name: 'Wheat Bread', price: 45)),
        SizedBox(width: 12),
        Expanded(child: ProductCard(name: 'Onion (1 kg)', price: 34, mrp: 40,
discount: '15% OFF')),
      ],
    ),
  ],
);
}

// ---- Floating cart bar ----
Widget buildFloatingCartBar() {
  return Container(
    padding: const EdgeInsets.symmetric(horizontal: 16, vertical: 12),
    decoration: BoxDecoration(
      color: const Color(0xFF1DA751),
      borderRadius: BorderRadius.circular(12),
    ),
    child: const Row(
      mainAxisAlignment: MainAxisAlignment.spaceBetween,
      children: [
        Text('4 items | Rs 154', style: TextStyle(color: Colors.white, fontWeight:
FontWeight.bold)),
        Text('View cart >', style: TextStyle(color: Colors.white, fontWeight:
FontWeight.bold)),
      ],
    ),
  );
}

// ---- Product card: Stack ----
class ProductCard extends StatelessWidget {
  final String name;
  final int price;
  final int? mrp;
  final String? discount;

  const ProductCard({super.key, required this.name, required this.price, this.mrp,
this.discount});

  @override
  Widget build(BuildContext context) {
    return Container(
      decoration: BoxDecoration(
        border: Border.all(color: Colors.grey.shade300),
        borderRadius: BorderRadius.circular(12),
      ),
      child: Column(
        crossAxisAlignment: CrossAxisAlignment.start,

```

```

        children: [
          Stack(
            children: [
              Container(
                height: 96,
                decoration: const BoxDecoration(
                  color: Color(0xFFFBEEE0),
                  borderRadius: BorderRadius.vertical(top: Radius.circular(12)),
                ),
              ),
              if (discount != null)
                Positioned(
                  top: 6,
                  left: 6,
                  child: Container(
                    padding: const EdgeInsets.symmetric(horizontal: 6, vertical: 2),
                    color: Colors.red,
                    child: Text(discount!, style: const TextStyle(color: Colors.white,
fontSize: 9)),
                  ),
                ),
              const Positioned(
                top: 6,
                right: 6,
                child: CircleAvatar(radius: 10, backgroundColor: Colors.white, child:
Icon(Icons.favorite_border, size: 11)),
              ),
            ],
          ),
          Padding(
            padding: const EdgeInsets.all(8),
            child: Column(
              crossAxisAlignment: CrossAxisAlignment.start,
              children: [
                Text(name, style: const TextStyle(fontSize: 12.5, fontWeight:
FontWeight.w600)),
                const SizedBox(height: 4),
                Row(
                  mainAxisAlignment: MainAxisAlignment.spaceBetween,
                  children: [
                    Text('Rs $price', style: const TextStyle(fontWeight:
FontWeight.bold)),
                    Container(
                      padding: const EdgeInsets.symmetric(horizontal: 10, vertical: 4),
                      decoration: BoxDecoration(color: const Color(0xFF6A0DAD),
borderRadius: BorderRadius.circular(6)),
                      child: const Text('ADD', style: TextStyle(color: Colors.white,
fontSize: 10)),
                    ),
                  ],
                ),
              ],
            ),
          ),
        ],
      );
    }
  }
}

```

5. Result — Screenshot of Output

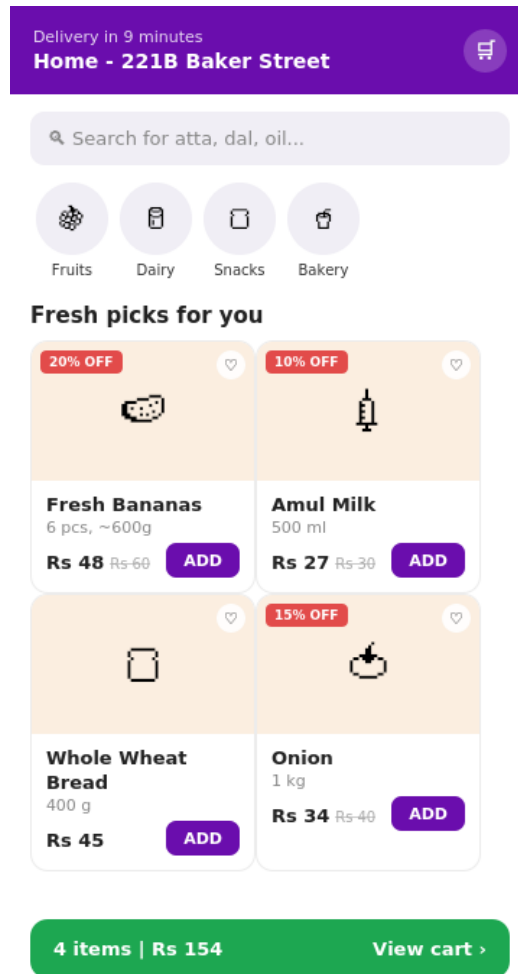


Fig. 1 — Grocery ordering app home screen built using Row, Column, and Stack widgets

6. Result

The grocery ordering app home screen was successfully built and run. The app bar and search bar (Row), the vertically arranged product grid (Column), and the overlapping badges and floating cart bar (Stack) were all displayed correctly, matching the intended design.

7. Conclusion

This mini project demonstrated how Row, Column, and Stack can be combined to build a realistic, professional-looking screen rather than isolated demo boxes. Row and Column were used together to organize the overall page structure and the product grid, while Stack was essential wherever elements needed to overlap — such as discount badges on product images and the floating cart bar staying fixed above the scrollable content. This shows that real-world Flutter UIs are built by nesting and combining these basic layout widgets rather than using them in isolation.